Practice workbook · Scale per-unit or reference quantities and combine them

A content per 100 g, per kg, per unit or per pot is given; amounts for another mass are found, combined, or used to find how much is needed.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-REFERENCE-RATIO-SCALING

1. Question 2 · part a · 3 marks

June 2023 Paper 2

Worked-example retry · Attempt part a only.

2 The table gives the nutritional information for a pot of yoghurt.

	Mass per 100 g of yoghurt	Mass per 85 g pot
fat	2.7 g	2.3 g
carbohydrate	12.6 g	10.7 g
protein	6.1 g	
calcium	149 mg	127 mg
vitamin D	0.9 µg	0.8 µg

(a) Calculate the mass of protein in one pot of yoghurt.

Give your answer to two significant figures.

(3)

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mass of protein = ... g

(b) Describe how the yoghurt could be tested to show that protein is present.

(3)

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1. Question 2 · part a · 3 marks · Working and answer

Working and answer

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2. Question 8 · part a · 4 marks

June 2019 Paper 1R

Further practice · Attempt part a only.

8 The table shows the rate of blood flow through different organs when a person is at rest. The table also shows the masses of these organs.

Organ	Mass in kg	Rate of blood flow in cm^3 per minute	
		through the whole organ	per 100g of the organ
kidneys	0.3	1200	
skeletal muscles	25.0		3
liver	1.5	1500	100

(a) Calculate the missing values and put them in the table.

(4)

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(b) Explain why the liver has a high rate of blood flow.

(4)

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2. Question 8 · part a · 4 marks · Working and answer

Working and answer

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3. Question 3 · part a.v · 4 marks

June 2024 Paper 1R

Further practice · Attempt part a.v only.

3 (a) The table gives the recommended daily dietary intake for different groups of males and females.

Group	Energy in kJ	Protein in g	Vitamin A in mg	Vitamin C in mg	Iron in mg
MALES					
15 to 18 years old	12 600	59	0.9	50	12
25 to 50 years old	12 200	63	0.9	60	10
FEMALES					
15 to 18 years old	9200	46	0.7	50	15
25 to 50 years old	9200	50	0.7	60	15

(i) Describe what is meant by a balanced diet.

(2)

(ii) Explain the difference between the amount of energy needed by males compared with females in the age range 25 to 50 years.

Use data from the table to help your answer.

(3)

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P 7 4 4 9 2 A 0 8 2 4

3. Question 3 · part a.v · 4 marks

June 2024 Paper 1R

Further practice · Attempt part a.v only.

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(iii) Explain why females in the age range 25 to 50 years need more iron in their diet than males in the same age range.

(2)

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(iv) State a nutrient, shown in the table, that can help to reduce night blindness.

(1)

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(v) Red meat and some green vegetables are good sources of iron.

100 g of red meat contains 2.5 mg of iron.

100 g of broccoli contains 0.70 mg of iron.

A male in the age range 15 to 18 years eats 250 g of red meat and 200 g of broccoli in one meal.

Calculate the extra mass, in mg, of iron the male needs to eat to meet his recommended daily dietary intake of iron.

(4)

extra mass = ... mg

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3. Question 3 · part a.v · 4 marks

June 2024 Paper 1R

Further practice · Attempt part a.v only.

(b) In the 19th century more than two million sailors on long ocean voyages died from a nutritional disease called scurvy.

(i) Give a symptom of scurvy.

(1)

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.....

(ii) Suggest why scurvy is an uncommon nutritional disease in the modern world.

(1)

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(Total for Question 3 = 14 marks)

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Working and answer

4. Question 5 · part e · 3 marks

June 2024 Paper 2R

Further practice · Attempt part e only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The heart is a double-pump that is part of the human circulatory system. It helps ensure that the blood and its components reach all cells in the body.

Some components of the blood are made in the bone marrow. An average adult with a mass of 70 kg has 5.5 dm^3 of blood. About 45% of the blood is made up of different cells. There are several different types of cell in the blood but red blood cells are the most common. In males, 1.0 dm^3 of blood contains 5.9×10^{12} red blood cells. In females the same volume of blood contains 5.5×10^{12} red blood cells. Haemoglobin makes up a large part of red blood cells. This protein is vital to the function of red blood cells.

Red blood cells are transported around the body in a liquid known as plasma. Only 10% of blood plasma is made up of components that are used or produced by body cells. The rest of the plasma is made up of water. Some components of blood plasma are used by body cells in chemical reactions, some act as messengers or have a role in blood clotting and others are excreted from the body as waste.

(a) Name the main tissue that makes up the walls of the heart. **(1)**

(b) Explain why the heart is known as a double-pump. (Lines 1 to 2) **(3)**

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P 7 4 4 9 5 A 0 1 4 2 8

4. Question 5 · part e · 3 marks

June 2024 Paper 2R

Further practice · Attempt part e only.

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(c) Red blood cells are the most common type of cell found in the blood.

Which of these are also types of cell found in the blood?

(1)

- ☒ **A** phagocytes and platelets
- ☒ **B** antibodies and hormones
- ☒ **C** lymphocytes and phagocytes
- ☒ **D** platelets and lymphocytes

(d) (i) Name a component of the blood plasma that acts as a messenger. (Lines 12 to 14)

(1)

.....

(ii) Name a component of the blood that is involved in blood clotting.

(1)

.....

(e) Calculate the number of red blood cells found in the blood of an adult male with a mass of 95 kg. (Lines 3 to 7)

Give your answer in standard form.

(3)

number of red blood cells = ...

(Total for Question 5 = 10 marks)

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4. Question 5 · part e · 3 marks · Working and answer

Working and answer

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5. Question 7 · part a.iii · 3 marks

June 2022 Paper 2

Further practice · Attempt part a.iii only.

7 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Our body is under constant attack from free radicals. Free radicals are molecules produced during chemical reactions that take place in body cells. Free radicals can change the structure and function of cells and damage genetic material. Some scientific evidence also suggests that free radicals can affect the flow of substances that enter and leave a cell.

Vitamin C is water-soluble and an important component of the daily diet. The recommended intake of vitamin C is 90 mg per day. Vitamin C helps to strengthen the immune system and improves the absorption of iron into the body. Vitamin C is also known as an antioxidant. Antioxidants protect cells from damage by free radicals and other substances by changing the way that free radicals behave.

Water and high temperatures can reduce the vitamin C content of food. For example, 40 g of fresh broccoli provides 80% of the recommended daily intake of vitamin C. The same mass of boiled broccoli provides 60% of the recommended daily intake of vitamin C. Boiling is the biggest cause of vitamin C loss, although baking and frying also lower vitamin C content.

(a) (i) Free radicals can damage genetic material in cells.

Describe one effect that this could have on body cells. (lines 2 and 3)

(2)

(ii) Suggest how antioxidants, such as vitamin C, allow cells to function normally. (lines 5 and 6 and lines 8 to 10)

(2)

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5. Question 7 · part a.iii · 3 marks

June 2022 Paper 2

Further practice · Attempt part a.iii only.

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(iii) Calculate the mass, in grams, of boiled broccoli that should be eaten to provide the recommended daily intake of vitamin C. (lines 12 to 14)

(3)

mass = ... g

(b) DCPIP is a chemical used to determine the amount of vitamin C in food.

Describe a method, using DCPIP, to compare the amount of vitamin C in fresh orange juice with that in boiled orange juice.

(5)

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(Total for Question 7 = 12 marks)

5. Question 7 · part a.iii · 3 marks · Working and answer

Working and answer

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6. Question 2 · part b.ii · 3 marks

November 2024 Paper 2

Further practice · Attempt part b.ii only.

2 (a) Describe what is meant by a balanced diet.

(2)

(b) A label on a pack of four slices of roast beef gives information about how much of each dietary component is present in the meat.

It also gives the percentage of the recommended daily amount for each dietary component.

Table 1 and table 2 give these values.

Energy in kJ	Fat in g	Sugars in g	Salt in g
470	2.4	less than 0.4	0.6

Table 1

Percentage of recommended daily amount (%)			
energy	**fat**	**sugars**	**salt**
4.0	4.0	less than 1.0	12.0

Table 2

(i) Name another dietary component that could be listed on the pack of meat.

(1)

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P 7 5 9 5 9 A 0 4 2 4

6. Question 2 · part b.ii · 3 marks

November 2024 Paper 2

Further practice · Attempt part b.ii only.

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(ii) Using data from the tables, calculate the mass of fat, in g, that is needed to achieve the recommended daily amount.

(3)

mass = ... g

(iii) Calculate the number of slices of meat a person would need to eat to supply a recommended daily energy intake of 11 800 kJ.

(2)

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number of slices = ...

6. Question 2 · part b.ii · 3 marks · Working and answer

Working and answer

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7. Question 2 · part c · 2 marks

June 2023 Paper 2

Further practice · Attempt part c only.

2 The table gives the nutritional information for a pot of yoghurt.

	Mass per 100 g of yoghurt	Mass per 85 g pot
fat	2.7 g	2.3 g
carbohydrate	12.6 g	10.7 g
protein	6.1 g	
calcium	149 mg	127 mg
vitamin D	0.9 µg	0.8 µg

(a) Calculate the mass of protein in one pot of yoghurt.

Give your answer to two significant figures.

(3)

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mass of protein = ... g

(b) Describe how the yoghurt could be tested to show that protein is present.

(3)

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7. Question 2 · part c · 2 marks

June 2023 Paper 2

Further practice · Attempt part c only.

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(c) A pot of yoghurt provides 16% of a person's daily requirement of calcium.

Calculate how many pots of yoghurt a person would need to eat to obtain their daily requirement of calcium.

(2)

number of pots = ...

(d) Explain why yoghurt is good for bone development.

(5)

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(Total for Question 2 = 13 marks)

7. Question 2 · part c · 2 marks · Working and answer

Working and answer

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8. Question 2 · part b.iii · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.iii only.

2 (a) Describe what is meant by a balanced diet.

(2)

(b) A label on a pack of four slices of roast beef gives information about how much of each dietary component is present in the meat.

It also gives the percentage of the recommended daily amount for each dietary component.

Table 1 and table 2 give these values.

Energy in kJ	Fat in g	Sugars in g	Salt in g
470	2.4	less than 0.4	0.6

Table 1

Percentage of recommended daily amount (%)			
energy	**fat**	**sugars**	**salt**
4.0	4.0	less than 1.0	12.0

Table 2

(i) Name another dietary component that could be listed on the pack of meat.

(1)

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P 7 5 9 5 9 A 0 4 2 4

8. Question 2 · part b.iii · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.iii only.

(ii) Using data from the tables, calculate the mass of fat, in g, that is needed to achieve the recommended daily amount.

(3)

mass = ... g

(iii) Calculate the number of slices of meat a person would need to eat to supply a recommended daily energy intake of 11 800 kJ.

(2)

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number of slices = ...

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8. Question 2 · part b.iii · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.iii only.

(iv) When meat is rubbed on a piece of paper it leaves a greasy mark due to the presence of fat in the meat.

Describe another test to show the presence of fat in the meat.

(5)

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(Total for Question 2 = 13 marks)

8. Question 2 · part b.iii · 2 marks · Working and answer

Working and answer

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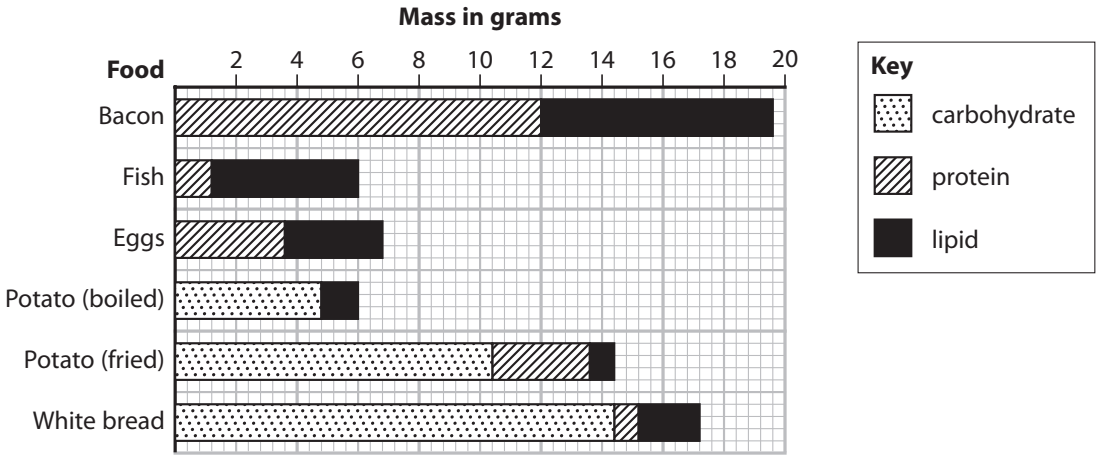
9. Question 7 · part a.iv · 3 marks

January 2022 Paper 1

Further practice · Attempt part a.iv only.

7 Most diets contain carbohydrates, lipids and proteins.

The bar chart shows the masses of each of these in 26 g samples of different foods.



(a) (i) State the names of the three chemical elements found in all carbohydrates, lipids and proteins.

(1)

(ii) State the name of one chemical element found in proteins but not found in carbohydrates or lipids.

(1)

(iii) Deduce the mass of lipid in 26 g of bacon.

(2)

mass of lipid = ... g

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P 6 7 0 5 9 A 0 2 0 2 4

9. Question 7 · part a.iv · 3 marks

January 2022 Paper 1

Further practice · Attempt part a.iv only.

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(iv) The energy in one gram of protein is 17 kJ and the energy in one gram of lipid is 37 kJ.

Calculate the total energy in 26 g of fish.

(3)

total energy = ... kJ

(b) Describe how to test for the presence of lipid in the fish.

(3)

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(Total for Question 7 = 10 marks)

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9. Question 7 · part a.iv · 3 marks · Working and answer

Working and answer

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10. Question 6 · part b.i · 3 marks

November 2020 Paper 2

Further practice · Attempt part b.i only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The effects of alcohol

Humans have been drinking alcohol for thousands of years. Drinking too much alcohol over many years can cause diabetes, malnutrition and diseases of the central nervous system and the liver. One immediate side effect of drinking too much alcohol is increased urination.

Beer is about 95% water and only 5% alcohol. The liver converts that 5% of alcohol into the same volume of water and some carbon dioxide. So if a person drinks 200 cm^3 of beer (one unit), the end result is 200 cm^3 of water. However, this person will not just urinate 200 cm^3 of urine, in fact the person urinates a total of 320 cm^3 of urine. So in general each unit of alcohol makes the person urinate an extra 120 cm^3 of urine more than the normal urine output.

A person with a mass of 60 kg produces 60 cm^3 of urine an hour and a person with a mass of 80 kg produces 80 cm^3 an hour and so on. Therefore if a 60 kg person drinks 200 cm^3 of beer in an hour they will produce 60 cm^3 of urine plus 320 cm^3, making a total of 380 cm^3.

Alcohol interferes with the mechanism that regulates water levels in our body because alcohol affects a small gland called the pituitary gland. One hormone released by the pituitary gland is anti-diuretic hormone (ADH). Alcohol reduces how much ADH is released by the pituitary. Each unit of alcohol that a person drinks forces the kidneys to produce an extra 120 cm^3 of urine more than the normal output.

Logic suggests that drinking lots of water would replace the 120 cm^3 of urine produced, but it is not that simple. The body only retains half the water that a person drinks. The other half leaves the body in urine, so the person will become dehydrated even if they drink lots of water.

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10. Question 6 · part b.i · 3 marks

November 2020 Paper 2

Further practice · Attempt part b.i only.

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(a) (i) Where in the body is the pituitary gland (line 16)?

(1)

- ☒ **A** brain
- ☒ **B** kidney
- ☒ **C** liver
- ☒ **D** pancreas

(ii) Which system includes the pituitary gland (line 16)?

(1)

- ☒ **A** digestive
- ☒ **B** endocrine
- ☒ **C** nervous
- ☒ **D** skeletal

(b) A person weighs 75 kg and drinks 400 cm^3 of beer in an hour (lines 11–14).

(i) Calculate the volume of urine produced by the person in this hour.

(3)

volume of urine = ... cm^3

(ii) Calculate the percentage increase in the volume of urine produced by this person in this hour.

(3)

percentage increase = ...%

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10. Question 6 · part b.i · 3 marks · Working and answer

Working and answer

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